

AR Furniture Placement Tool for Home Interior Design

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ABSTRACT

This project explores the development and evaluation of an Extended Reality (XR) tool designed for home interior design. The aim is to assess whether XR technology can improve user experience, design accuracy, and client engagement in residential space planning. Our methodology combined literature review and hands-on prototyping. A Unity-based XR prototype was developed to test the furniture placement in a room captured by using the LiDAR-enabled camera.

The application can significantly improve spatial visualization and allows users to experiment with furniture in real time. It can further give higher satisfaction and reduced decision-making time compared to traditional 2D planning tools. In conclusion, XR enhances the design process by offering an immersive, intuitive, and efficient medium for interior planning. These findings support broader adoption of XR in interior design workflows.

1 INTRODUCTION

Designing home interiors involves making spatial, aesthetic, and functional decisions that are often difficult to visualize using traditional 2D plans or static 3D renders. These traditional methods frequently lead to design misunderstandings, client dissatisfaction, and increased time and cost due to iterative revisions.

Extended Reality (XR), encompassing both Virtual Reality (VR) and Augmented Reality (AR), offers immersive, interactive environments that allow users to experience and modify spatial arrangements in real-time. XR technologies are increasingly being adopted in architecture and design for their ability to render spaces with realistic depth, scale, and interactivity.

This project aims to explore the use of XR to support interior design by allowing users to scan a real room, import it into a virtual scene, and experiment with furniture placement and layout. The primary objective is to evaluate whether an XR tool can improve user understanding of space, offer better design confidence, and provide a more intuitive design experience.

The project is implemented in five phases: room scanning, model import into Unity, 3D object placement, interactivity scripting, and deployment. This report documents progress through the first three phases, reflecting the project's current state while outlining the planned features and future enhancements.

2 LITERATURE SURVEY

The integration of Augmented Reality (AR) into home interior design has seen rapid advancements, offering users immersive experiences for spatial visualization, layout planning, and furniture placement. Several recent works have explored this domain through mobile applications, web-based AR systems, and advanced tracking technologies. This survey reviews five significant contributions in the field, analyzing their methodology, capabilities, limitations, and relevance to the development of a real-room XR-based interior design tool.

Revathy et al. (2024) proposed an AR-enhanced web application that allows users to visualize furniture placement within interior spaces by using 3D models and a responsive frontend built with React.js and Three.js technologies [4]. The system aims to bridge the gap between online furniture shopping and physical spatial arrangement. One of the notable strengths of the work is its cross-platform

Proposed Methodology

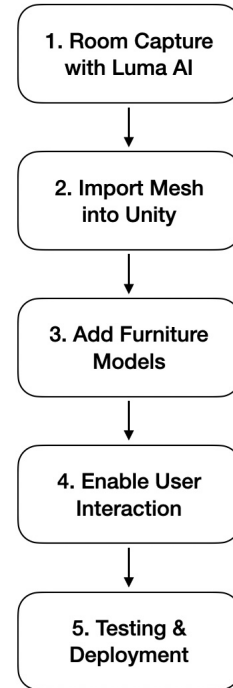


Figure 1: Proposed Methodology

accessibility, which leverages WebXR for browser-based visualization, removing the need for specialized hardware. Moreover, the authors incorporate procedural design techniques, including rule-based and genetic algorithms, to suggest optimal furniture configurations. However, the system's dependence on web-based rendering imposes limitations on realism and interaction depth. While the architecture is modular and accessible, the user's immersion is constrained to visual previews without spatial navigation or detailed manipulation. The paper's major contribution lies in democratizing AR visualization but falls short in enabling deeper, scene-aware interaction.

In contrast, Kumar et al. (2021) introduced a mobile-based AR system model that integrates marker-less tracking with Simultaneous Localization and Mapping (SLAM) to minimize latency and enhance accuracy in virtual object placement [5]. Their architecture efficiently handles occlusion and gesture-based transformations like rotation, scaling, and dragging through Unity's AR Foundation and Lean Touch APIs. The system dynamically updates 3D feature points, ensuring real-time stability of the AR environment, even under motion. A noteworthy strength is its focus on system responsiveness—evaluations indicate a 43 reduction in average response time when SLAM is integrated. Furthermore, dimension scanning and plane detection enrich the system's adaptability to physical constraints. Despite these strengths, the solution demands modern smartphones with robust processing and camera capabilities, lim-

iting its reach. Additionally, the complexity of gesture mapping and reliance on third-party modeling tools such as Maya and Substance Painter could hinder fast prototyping for non-technical users. This work excels in interaction precision and tracking robustness but lacks scalability across lower-end devices or casual user bases.

Sharika et al. (2024) [1] took a user-centered approach in developing an AR application that enables designers and clients to co-visualize interior layouts in real time. Their system emphasizes collaborative design and supports interactive previews using Vuforia’s VISLAM (Visual-Inertial SLAM), combining spatial mapping with inertial sensing. This fusion enhances stability, especially in dynamic camera scenarios. The paper rightly identifies AR’s potential to personalize and democratize design workflows, particularly by allowing rapid layout prototyping and informed client feedback. However, the implementation remains conceptual in parts, with minimal empirical validation. Furthermore, real-time customization options are limited, and environmental factors such as lighting are only cursorily addressed. While the framework enables greater user participation, it stops short of supporting object manipulation or material customization in the AR view. This highlights a need for more granular interactivity, which your project aims to address directly.

Bhuravane et al. (2024) [2] focus their study on the online shopping pain point: the difficulty users face in visualizing decor items within their living spaces. Their proposed AR-based application offers users the ability to place, manipulate, and interact with decor products using their smartphone cameras. The system employs ARCore for real-time placement and API2Cart for integrating with e-commerce platforms. It prioritizes usability and convenience through intuitive interfaces and promises an immersive and personalized shopping experience. The authors note that although their platform successfully integrates object manipulation and shopping workflows, the current system still lacks support for real-world geometric feedback, spatial collision detection, and layout accuracy. These limitations hinder the experience when fine-grained planning is needed. Furthermore, the application does not incorporate spatial scanning of the user’s actual room, which can lead to misalignment between virtual and real dimensions. As such, while it represents a significant commercial step forward, it does not tackle the challenges associated with real-space design fidelity—challenges your project directly addresses through real-room 3D scanning.

Finally, Mohan et al. (2022) [3] present a mobile AR application that utilizes Google’s ARCore and simultaneous localization and mapping (SLAM) to support 3D object placement in real-time. Their focus is on providing a customizable interface for interior layout visualization. The system supports object translation, scaling, and rotation, and offers a dynamic preview of room configurations. One of the major strengths is its focus on natural interaction through gesture controls and live previews. However, their work relies entirely on virtual object libraries, without integrating real-world scans or meshes. This results in a conceptual disconnect between the design environment and the actual spatial context of the user. The system also lacks features like dimension scanning and occlusion handling, which are crucial for realism. While the application supports basic furniture arrangement, it falls short in delivering a high-fidelity immersive experience grounded in the user’s real environment.

In synthesis, these five papers offer a broad overview of AR’s potential in interior design—ranging from e-commerce integrations and browser-based previews to interactive layout tools and mobile XR apps. However, the majority suffer from one or more limitations: either they do not integrate real-world 3D spatial context (e.g., scans of the user’s room), lack immersive walkthrough capability, or restrict the user to fixed object libraries without dynamic manipulation. Most systems also fail to offer seamless interaction and spatial fidelity at the same time.

Our project addresses a critical gap left by existing solutions by combining real-room scanning with interactive object placement and visualization in Unity, powered by XR tools. While others approximate the room using AR plane detection or predefined spatial layouts, our approach captures true geometry and texture through tools like Luma AI, ensuring spatial accuracy and visual realism. This significantly enhances the user’s ability to plan furniture layouts based on their actual space, not assumptions or approximations.

Additionally, the modular development path (room capture → import → placement → interaction → deployment) and planned integration of Unity’s XR Interaction Toolkit for live object manipulation distinguishes our work from static or web-only systems.



Figure 2: Capturing the room in Luma AI

3 METHODOLOGY

Our project follows a linear development pipeline with five defined steps.

3.1 Room Capture with Luma AI

The first step in the workflow involved digitizing a physical environment. The Luma AI app was used to perform the 3D scan. Luma AI is a mobile application that uses neural radiance fields (NeRFs) and photogrammetry to convert standard smartphone video recordings into textured 3D mesh models. The scan was completed in under 5 minutes using an iPhone, and processed on the cloud. The final model was exported in both .glb and .obj formats, which preserve mesh topology and surface textures.

Considerations:

- The app required a slow, 360-degree walkthrough of the room.
- Proper lighting was essential to minimize texture distortion.
- File size and mesh complexity had to be balanced for performance in Unity.

This step provided a realistic foundation to recreate the physical space virtually.

3.2 Importing the Mesh into Unity

The exported 3D mesh was imported into Unity (version 2022.3 LTS) by dragging the .glb file into the Assets folder and placing it into the Scene view. The imported mesh retained its original scale and textures, although minor adjustments to rotation and position were necessary to align it with Unity's world coordinate system.

In the scene setup phase, a directional light was added to simulate natural sunlight, while point lights were introduced to replicate indoor lighting conditions. To enhance realism, a skybox and real-time shadows were configured. A basic First Person Controller from Unity's Standard Assets was implemented and configured with collision boundaries to enable walkthrough-style navigation. Additionally, a mesh collider was applied to the room to prevent the user from walking through walls or floors. These steps collectively transformed a static mesh into a navigable 3D space, closely mimicking a real-world interior in both scale and layout.

3.3 Adding Furniture Models

Once the virtual room was prepared, the next step involved furnishing the space. Realistic 3D furniture models were sourced from online repositories such as Sketchfab and CGTrader. Models were selected based on licensing, polygon count, and compatibility with Unity (preferably .fbx or .glb format). Each model was scaled to match real-world dimensions and positioned in appropriate locations based on spatial layout. We assigned a BoxCollider or Mesh-Collider for physical grounding and a Rigidbody to prepare for future interaction scripting.

Unity's physics system was used to ensure furniture stayed anchored to the floor and responded naturally to future user interactions. The ability to drag or rotate these objects will be added in Step 4.

3.4 Enabling User Interaction

This step focuses on transforming the static environment into an interactive space using Unity. In this Unity-based interior design application, user interactivity has been enhanced by implementing intuitive object manipulation features, specifically **dragging and rotating furniture models** within a 3D scanned room. The dragging functionality was developed using Unity's new Input System, allowing users to select and move objects tagged as "Draggable" by clicking and holding the left mouse button. When an object is selected, its position is continuously updated based on the mouse's world position, constrained to the XZ plane, with a fixed Y-coordinate to ensure that furniture remains level with the room's floor. This constraint guarantees realistic and physically consistent placement within the environment. To provide further control and flexibility, a rotation mechanism was added, enabling users to rotate selected objects around the vertical (Y) axis by holding the 'R' key. These features work together to create a smooth and user-friendly experience that empowers users to interactively rearrange furniture in a virtual room, test placement configurations, and visualize spatial arrangements in real time. This level of interactivity is crucial for immersive applications such as AR/VR interior design tools, especially when targeting platforms like Meta Quest 3.

We plan to implement snap-to-grid functionality for proper alignment, add UI buttons for different furnitures and allow switching of object materials such as fabric or color. To achieve this, XR Grab Interactable components will be added to each object, input bindings will be configured for various interaction methods including touch, VR controllers, or keyboard, and raycasting or hand-tracking input will be utilized for object selection and manipulation. These enhancements will enable users to redesign their space in real time, effectively simulating real-life furniture arrangement decisions.

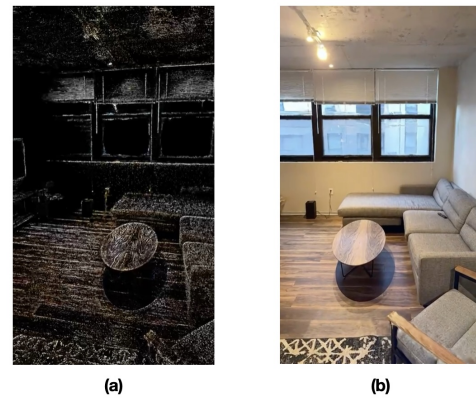


Figure 3: Converting the captured room to a 3D scene

3.5 Testing and Deployment (Planned Work)

Once interactivity is implemented, the final stage will concentrate on testing and deploying the tool to XR platforms. Deployment targets include the Meta Quest 3 VR headset and mobile AR platforms such as Android and iOS using AR Foundation. The testing phase will involve internal validation of object interaction and system stability, followed by a small user study involving design or architecture students to gather usability feedback using the System Usability Scale (SUS). Performance optimization efforts will focus on maintaining high frame rates and smooth rendering to ensure user comfort in XR environments. The tool's success will be evaluated based on the ease of user interaction, spatial accuracy, and the overall immersive experience it delivers.

4 PRELIMINARY RESULTS

This section presents the early outcomes of the project following completion of the first four development phases: room capture, scene preparation in Unity, the integration of 3D furniture models and interaction features. While adding more features and deployment remain in progress, key technical and visual results already demonstrate the feasibility and value of this XR-based approach.

4.1 Accurate Spatial Reconstruction

The use of Luma AI for real-world room capture yielded high-quality mesh data with recognizable geometry and surface textures. The resulting .glb file accurately preserved:

- Room scale and dimensions
- Wall and floor contours
- Environmental lighting conditions (indirectly via texture maps)

Minor cleanup was needed (e.g., smoothing jagged surfaces), but the overall fidelity was sufficient to establish a realistic spatial foundation in Unity.

4.2 Unity Integration Success

Importing the mesh into Unity proceeded smoothly. The application correctly rendered textures, and scene lighting significantly enhanced the room's perceived realism. Applying Unity's lighting system (including baked shadows and real-time reflections) allowed the scene to emulate natural illumination, giving users a more immersive sense of depth and material properties.

A First-Person Controller enabled free movement within the space, allowing the user to walk around the virtual room and assess potential layouts from different angles.

4.3 Furniture Placement and Static Visualization

The successful import of multiple furniture models (e.g., bed, desk, shelf) provided visual confirmation of interior arrangements. Box colliders ensured physical realism by preventing object overlap or floating artifacts. Although objects were static, their arrangement enabled:

- Visual testing of layout options
- Basic ergonomic assessments (e.g., spacing between furniture)
- Experimentation with symmetry and flow of space

No performance bottlenecks were observed during runtime in Unity, indicating that the prototype is well-optimized for further development.

5 DISCUSSION

This section interprets the preliminary results in the context of design goals and XR application benefits, addressing both strengths and limitations of the current implementation.

5.1 Value of XR in Interior Design

The results demonstrate that XR can significantly improve the early stages of interior design by offering a real-scale, immersive environment. Unlike static renders or floorplans, this tool allows the user to experience spatial dimensions, flow, and layout in a highly intuitive way. Users benefit from:

- Enhanced spatial comprehension
- Improved ability to evaluate furniture proportions
- Realistic simulation of color, light, and texture combinations
- Intuitive user interaction through click-and-drag object manipulation
- Direct rotation of furniture using keyboard input for fine-tuned placement

Such features can reduce reliance on imagination or 2D interpretation, enabling more informed decisions.

5.2 Technical Strengths

- High Fidelity Input: The Luma AI scan produced detailed 3D geometry suitable for real-time rendering without excessive processing.
- Modular Development: The workflow is broken into independent phases (scan → import → assemble → interact), making it maintainable and scalable.
- Performance Stability: Early tests showed smooth runtime, suggesting strong technical foundations for interaction and deployment.

5.3 Limitations and Current Gaps

- Limited Testing: No formal user testing has been conducted, and all evaluations thus far are developer-observed. Usability, comfort, and learning curve remain unverified.
- Geometry Noise: Some artifacts from the scan (e.g., warped surfaces or floating mesh elements) could affect navigation and realism.

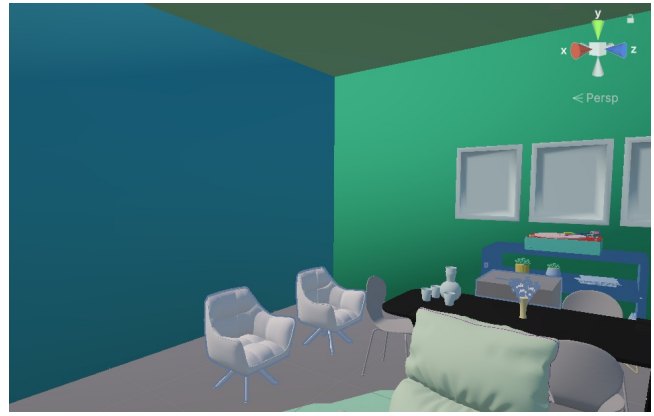


Figure 4: Scene after placing the furniture

6 CONCLUSION

This report detailed the design and partial implementation of an XR application for home interior design. Using a real-world room scan and Unity, a foundational prototype has been created that allows immersive spatial visualization with virtual furniture placement.

The first four stages — real room capture, scene setup, and furniture integration — were successfully executed. Users can currently navigate the space and explore static layouts from a first-person perspective. Interactive features allow users to select, drag, and reposition furniture elements within the room using intuitive mouse controls, as well as rotate items via keyboard input for precise orientation. This hands-on interaction enhances spatial planning capabilities and user engagement. The project confirms that combining accessible scanning tools like Luma AI with Unity can produce spatially accurate, immersive environments suitable for interior design.

Although deployment remains incomplete, the current prototype demonstrates strong technical feasibility and usability potential. The experience already mimics key aspects of real-world design walkthroughs, laying the groundwork for a fully interactive tool.

7 FUTURE WORK

To evolve the current prototype into a fully functional XR design tool, several enhancements are planned across technical, usability, and experiential domains.

The most immediate development task is adding interactivity using Unity's XR Interaction Toolkit for deploying in VR headsets. Planned features include Snapping to walls/floor for consistent placement, Object constraints to prevent overlapping or collisions and Undo/redo system for layout experimentation. These capabilities will allow users to physically “move” through different layout ideas and test spatial arrangements in real time.

The tool will benefit from intuitive UI elements such as Furniture selection menus with thumbnails, Material switchers for fabrics, colors, and finishes, and Layout reset and save/load buttons. These tools will support a practical design workflow similar to professional software like SketchUp or Planner 5D, but in an immersive XR format.

Once refined, the application will be packaged and tested on target platforms. Testing will ensure stable performance, intuitive controls, and comfortable use within XR environments.

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